

AeroGuard-Bob

HUMS Mission-Readiness & Predictive Maintenance Copilot

Transforming raw HUMS telemetry and maintenance logs into explainable readiness decisions and optimized repair schedules.

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Operational Friction in Military Aviation Maintenance

\$90B+ Annual Spend

Rigid Calendar Schedules

Fixed maintenance intervals miss condition-based warning signals, causing unexpected in-flight aborts and degraded readiness.

Telemetry Overload

Unanalyzed Sensor Feeds

Flight-line officers are inundated by unanalyzed HUMS telemetry and siloed historical maintenance logs without real-time triage.

Resource Bottlenecks

Grounded Jet Delays

Under tight technician crew and spare-parts constraints, commanders lack automated tools to decide which grounded aircraft to repair first.

AeroGuard-Bob: Predict -> Explain -> Prioritize -> Act

1. Data Fusion

Ingests real-time HUMS telemetry & 2-year maintenance records across 25 jets.

2. Readiness Triage

Classifies READY vs NON-READY with plain-language operational diagnostics.

3. Risk Scoring

Computes 0-100 failure risk scores across 5 critical subsystems.

4. Resource Optimizer

Greedily schedules repairs under crew and spare-parts availability constraints.

End-to-End Modular Architecture

Readiness Classifier

`src/
readiness_classifier.py`

Evaluates vibration, temperature & oil thresholds to classify READY vs NON-READY assets.

Subsystem Risk Engine

`src/
failure_predictor.py`

Computes 0-100 risk across Engine, Hydraulics, Fuel, Landing Gear, and Avionics.

Maintenance Planner

`src/
maintenance_planner.py`

Allocates crews and spare parts greedily while logging bottleneck causes for queued jets.

IBM Bob MCP Server

`src/mcp_server.py`

Exposes stdio JSON-RPC 2.0 tools for natural language queries in IBM Bob Copilot.

Conversational Decision Support via Model Context Protocol

MCP Tool Name	Capabilities & Description
<code>get_fleet_readiness</code>	Returns fleet-wide readiness metrics, total jets, READY count, and readiness percentage.
<code>get_asset_risk</code>	Provides asset-specific 0-100 subsystem risk scores and sensor telemetry diagnostics.
<code>get_maintenance_plan</code>	Generates constrained repair schedules based on available technician crews and spare parts.

Natural Language Command Center

- Commanders query squadron status directly in IBM Bob using plain English.
- MCP standardizes tool invocation via JSON-RPC 2.0 stdio stream.
- Every repair plan is grounded with audit-ready constraint logs and bottleneck explanations.

Technology Stack & Key Highlights

Tech Stack Summary

Language: Python 3.10+

Protocol: Model Context Protocol (MCP) over JSON-RPC 2.0 stdio

AI / Assistant: IBM Bob Copilot integration

Data Formats: CSV telemetry & 2-year maintenance logs

Logic Engines: Explainable rule-based scoring & greedy resource optimizer

What We're Most Proud Of

- ✓ End-to-End Decision Pipeline: Seamless integration from sensor telemetry ingestion to conversational actuation in IBM Bob.
- ✓ Auditable Explainability: Plain-language diagnostic reasons instead of black-box prediction outputs.
- ✓ Resource-Aware Optimization: Real-world technician and spare-parts stockout constraint modeling.

Honest Prototype Limitations & Future Roadmap

Prototype Limitations

- Synthetic Telemetry: Currently tested on simulated HUMS data across 25 aircraft rather than live flight feeds.
- Rule-Based Failure Engine: Uses explainable threshold scoring pending large-scale validation on real military HUMS datasets.
- Batch Ingestion: Real-time streaming pipeline scaffolded; currently processes batched CSV telemetry files.

Future Roadmap

- Live HUMS Ingestion: Connect directly to aircraft telemetry streaming endpoints and IoT gateways.
- Enterprise Security & Auth: Add role-based access control (RBAC), multi-factor authentication, and audit trails.
- Multi-Squadron Logistics: Expand optimizer to manage spare-parts supply chains across regional airbases.